ALEJANDRO SÁNCHEZ

Data Scientist | Data Analyst | AI Developer

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After four years working as a bioinformatics and biotechnology researcher—analyzing complex biological data in international centers—I transitioned to Artificial Intelligence to build tangible, predictive solutions. Drawing on my background, my profile combines analytical and scientific rigor with modern software development: I design Machine Learning models and intelligent AI applications to solve real-world problems

PROFESSIONAL EXPERIENCE

05.2025 – 12.2025
Madrid, Spain
(Online)

Data Scientist (Capstone Project: DSMarket) - *Nuclio Digital School*

- Developed an end-to-end retail demand forecasting system, processing 5 years of historical transaction data: 10 stores, >3,000 products, >500k records.
- Developed a clustering algorithm (**K-Means**) for product segmentation to optimize inventory.
- Trained and deployed **Machine Learning** time series models (XGBoost, Random Forest) and Lasso Regression for demand forecasting, integrating them into an interactive Power BI dashboard for executive decision-making.

09.2022 – 07.2025
Madrid, Spain

Bioinformatician & Data Scientist - *Centro de Biología Molecular Severo Ochoa (CSIC-UAM)*

- Led the computational strategy for the research group's multi-omics projects, generating insights from raw biological data.
- Built and optimized **ETL** pipelines in **Python, Bash, and R**, significantly reducing processing times.
- Beyond technical execution, managed supplier budgets, mentored 5 junior researchers, and presented complex statistical data to non-technical stakeholders.
- Co-authored 2 peer-reviewed scientific publications as first and second autor.

03.2025 – 06.2025
Online

Data Analyst (AI Evaluation & RLHF) - *Telus International AI (Freelance)*

- Evaluated data quality for search algorithms and computer vision models using Reinforcement Learning from Human Feedback (RLHF).
- Reviewed and generated strict "Ground Truth" data based on complex guidelines to refine and improve the accuracy of commercial AI models.

04.2020 – 01.2022
Konstanz, Germany

Researcher - Bioinformatics and Biotech - *Universität Konstanz (AG Kroth)*

- Managed end-to-end data pipelines for environmental transcriptomics projects, from experimental design, processing, and quality control to final results extraction.
- Conducted multivariate statistical analysis in R and de novo assembly pipelines in Bash, utilizing remote High-Performance Computing (HPC) environments.
- Mentored 4 international students in experimental design and advanced data analysis.

EDUCATION

05.2025 – 12.2025
Online

Master's Degree in Data Science & AI (Grade: 9.0/10)
Nuclio Digital School

09.2019 – 07.2022
Barcelona (Online)

Master's Degree in Bioinformatics and Biostatistics (Grade: 8.5/10)
Universitat Oberta de Catalunya (UOC)
Master's Thesis: Differential analysis of dinoflagellate transcriptomes. Development of an end-to-end pipeline.

09.2017 – 09.2018
Madrid, Spain

Master's Degree in Microbiology (Grade: 8.5/10)
Universidad Autónoma de Madrid (UAM)
Master's Thesis: Nucleotide insertion of a bacterial DNA polymerase (scientific publication in Biomolecules)

09.2013 – 07.2017
Salamanca, Spain

Bachelor's Degree in Biology (Grade: 8.0/10)
Universidad de Salamanca

CERTIFICATIONS

Data Analytics Job Simulation
Deloitte Forage
Power BI *(Santander Open Academy)*
ICH Good Clinical Practice (GPC)
GDPR for Healthcare *(Alison)*
Big Data in Transportation
(Fundacion Maude)
Cisco Data Analytics Essentials

TECHNICAL STACK

Data Science & ML
Python (Pandas, NumPy, Scikit-learn, TensorFlow), R, Time Series, Clustering (K-Means), Statistical Validation, A/B Testing

Business Intelligence
Power BI, DAX, Tableau, Streamlit, R shiny dashboards

Data Engineering & Cloud
SQL, Bash Scripting, Linux, Git, Docker, Pipelines ETL, HPC (Slurm), GNU Parallel. AI, AI orchestration, LLMs, NLP

Domain Expertise
Bioinformatics, NGS (Illumina, ONT), Transcriptomics, CRISPR-Cas9, Bioreactors, Cell Cultures

LANGUAGES

SPANISH (Native) **| ENGLISH** (C1 - Cambridge CAE) **| GERMAN** (B1 - Intermediate)